Digital Signal Processing – Assignment 1
Fourier Transform: Voice Analysis, Enhancement and Aural Exciter

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1 Recordings

Voice of one team member, one sentence: "*Saad and Will are electro wizards.*" (language: English). Both recordings are uncompressed WAV, 48 kHz, 24-bit PCM, mono, not clipped, at distances of 5 cm and 1 m.

2 Task 1: Time and Frequency Analysis (20%)

2.1 Time and frequency domain plots

Placeholder: time + frequency plot, 5 cm

Figure 1: Voice recorded at 5 cm: (left) time domain, (right) frequency domain.

Placeholder: time + frequency plot, 1 m

Figure 2: Voice recorded at 1 m: (left) time domain, (right) frequency domain.

2.2 Vowel fundamentals

Table 1: Visually identified vowel fundamentals.

Vowel	f_0 at 5 cm (Hz)	f_0 at 1 m (Hz)	Justification
/a/	–	–	–
/e/	–	–	–
/o/	–	–	–

2.3 Consonant range

2.4 Noise region

Placeholder: annotated spectra (vowel peaks, consonant band, noise region)

Figure 3: Annotated spectra: vowel fundamentals, consonant range and noise-only region.

3 Task 2: Speech Enhancement (20%)

3.1 Method

3.2 Noise to be removed

3.3 Bass loss with distance and pop sounds

Table 2: Gain adjustments applied (final settings).

Recording	Band (Hz)	Gain (dB)	Purpose	Reason
5 cm	0–60	-12	Low cut	Pops / rumble
5 cm	95–400	+3.5	Bass boost	Depth
5 cm	3000–10000	+6	Brightness	Clarity
5 cm	12200–Nyq	-12	High cut	Hiss
1 m	–	–	–	–

Placeholder: gain curves (5 cm and 1 m)

Figure 4: Gain control curves applied to each recording.

Placeholder: overlay of enhanced 5 cm vs 1 m spectra

Figure 5: Enhanced spectra, 5 cm vs 1 m.

4 Task 3: Aural Exciter (60%)

4.1 Design overview

Placeholder: block diagram of your exciter

Figure 6: Block diagram of the implemented exciter.

4.2 Pre-processing

4.3 Iterative development

4.3.1 Iteration 1: initial design

4.3.2 Iteration 2: ...

4.3.3 Iteration 3: ...

Placeholder: annotated spectrum 2 kHz to 12 kHz, before vs after

Figure 7: Annotated spectrum (2 kHz to 12 kHz) before and after the exciter, showing the generated harmonics near 6 kHz.

Placeholder: annotated comparison of drive levels / design changes

Figure 8: Effect of drive level (or design change) on the harmonic content.

4.4 Final design and justification

Table 3: Final exciter parameters.

Parameter	Value
Sidechain band (Hz)	–
Non-linearity / drive	tanh, drive = –
Harmonic selection band (Hz)	–
Mix ratio	–
Big-bottom band (Hz) (if used)	–

Acknowledgement of GenAI Use

Tool: Claude (version: ..., publisher: Anthropic).
Use: ...
Own contribution: ...

A Python Code

A.1 run_tasks.py

A.2 task1_plot.py

A.3 task2_enhance.py

A.4 task3_exciter.py